⛽ Gas Fee Optimizer & Batch Transaction System

KRITI 2026 • Web3Assam • Submission Document

Deployed UI

Live Frontend (GitHub Pages):

https://ask-812.github.io/Web3/

Relayer Service

Live Relayer API (Vercel Serverless):

https://relayer-sigma.vercel.app

GitHub Repository

Source Code:

https://github.com/Ask-812/Web3

Network: Ethereum Sepolia Testnet

Deployed Contract Addresses

Contract	Address
BatchExecutor	`0x2c61E8959b5602C1DD9399fB16ABE9182fc7E3E7`

CompressedBatchExecutor	`0x01eAA34feF31F33E5EB15C5084569C1399B9C0CB`
SampleDApp	`0x47FF1aCB49cf23C28AA5887FC3bb5073D566f938`
SampleDAppMeta	`0xAE3329Cec2A4827dd13E9b86874CF175179B0E54`
GasSponsor	`0x856371cEED89a9c14E038e9cd3BF7026B004C663`
Forwarder	`0x21a94492478f4A00d29774c3754Ca8aB24FB8c9f`
GaslessToken	`0x66Eb4Fe980a960FC04553591c156e5DeDb9d7809`

Live Test Results (Sepolia, 5-Call Benchmark)

Execution Mode	Gas Used	Savings
🐌 Individual (5 txs)	822,578 gas	— (baseline)
📦 Standard Batch (1 tx)	776,980 gas	5.5%
⚡ Compressed Batch (1 tx)	765,597 gas	6.9%
🔒 Meta-Tx Relayer (gasless)	695,423 gas	**15.5% — user paid ZERO**

Key Features

Transaction Batching · Meta-Transactions (EIP-712) · Gas Sponsorship · Calldata Compression · Cross-User Bundling · EIP-2771 Context

Tech Stack

Solidity 0.8.24 • Hardhat • ethers.js • OpenZeppelin • Node.js Relayer • Vercel Serverless • GitHub Pages

How to Use the Live UI

1. Visit https://ask-812.github.io/Web3/
2. Connect MetaMask (auto-switches to Sepolia)
3. Click "Load Contracts" to load deployed contracts
4. Click "Quick Demo (5 Actions)" to load sample actions

5. Try all 4 execution modes and compare gas savings

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15+ smart contracts • 67+ tests • 6 EIP standards • Deployed on Sepolia